

LECTURE: FOURIER COSINE SERIES

1. FOURIER COSINE SERIES

Question: What if you want to write f as a cosine series?

Goal:

Find A_m such that $f(x) = \sum_{m=0}^{\infty} A_m \cos(\pi m x)$ on $(0, 1)$

Carefully note: We're starting with $m = 0$ here

Good news: Everything we said about sin is also true about cos and in particular, we get pretty much the exact same formula

$$\begin{aligned} A_m &= \frac{f \cdot \cos(\pi m x)}{\cos(\pi m x) \cdot \cos(\pi m x)} = \frac{\int_0^1 f(x) \cos(\pi m x) dx}{\int_0^1 \cos^2(\pi m x) dx} = \frac{\int_0^1 f(x) \cos(\pi m x) dx}{\frac{1}{2}} \\ &= 2 \int_0^1 f(x) \cos(\pi m x) dx \quad m = 1, 2, \dots \end{aligned}$$

Special Case: ($m = 0$) A_0 corresponds to $\cos(0x)$ and so

$$A_0 = \frac{f \cdot \cos(0x)}{\cos(0x) \cdot \cos(0x)} = \frac{\int_0^1 f(x) \cos(0x) dx}{\int_0^1 \cos^2(0x) dx} = \frac{\int_0^1 f(x) dx}{\int_0^1 1 dx} = 1 \int_0^1 f(x) dx \text{ (half!)}$$

Fourier Cosine Series:

If $f(x) = \sum_{m=0}^{\infty} A_m \cos\left(\frac{\pi m x}{L}\right)$ on $(0, L)$ then

$$A_0 = \frac{1}{L} \int_0^L f(x) dx \text{ and } A_m = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{\pi m x}{L}\right) dx$$

2. TABULAR INTEGRATION**Video:** Tabular Integration**Example 1:**

Find A_m such that $x^3 = \sum_{m=0}^{\infty} A_m \cos(\pi m x)$ on $(0, 1)$

Special Case: Always isolate the case $m = 0$

$$A_0 = \frac{1}{1} \int_0^1 x^3 dx = \frac{1}{4}$$

General Case: And for $m = 1, 2, \dots$ we have

$$A_m = \frac{2}{1} \int_0^1 x^3 \cos(\pi m x) dx$$

Trick: Here is cool trick for finding Fourier series of polynomials**STEP 1:** Put x^3 on the left hand side and $\cos(\pi m x)$ on the right**STEP 2:** Differentiate x^3 until you get 0 and integrate $\cos(\pi m x)$

STEP 3: Then cross multiply and alternate signs

$+$	x^3	$\cos(\pi mx)$
$-$	$3x^2$	$\sin(\pi mx)/\pi m$
$+$	$6x$	$-\cos(\pi mx)/(\pi m)^2$
$-$	6	$-\sin(\pi mx)/(\pi m)^3$
$+$	0	$\cos(\pi mx)/(\pi m)^4$

Then you get

$$\begin{aligned}
 & A_m \\
 &= 2 \left[\overset{+}{x^3} \left(\frac{\sin(\pi m x)}{\pi m} \right) \overset{-}{3x^2} \left(-\frac{\cos(\pi m x)}{(\pi m)^2} \right) \overset{+}{6x} \left(-\frac{\sin(\pi m x)}{(\pi m)^3} \right) \overset{-}{6} \left(\frac{\cos(\pi m x)}{(\pi m)^4} \right) \right]_0^1 \\
 &= 2 \left[0 - 3(1)^2 \left(-\frac{\cos(\pi m)}{(\pi m)^2} \right) + 0 - \left(6 \frac{\cos(\pi m)}{(\pi m)^4} \right) - 0 + 0 - 0 + 6 \left(\frac{\cos(0)}{(\pi m)^4} \right) \right] \\
 &= 6 \frac{(-1)^m}{(\pi m)^2} - 12 \frac{(-1)^m}{(\pi m)^4} + \frac{12}{(\pi m)^4} \\
 &= \frac{6}{(\pi m)^2} (-1)^m + \frac{12}{(\pi m)^4} \underbrace{((-1)^{m+1} + 1)}_{0 \text{ or } 2}
 \end{aligned}$$

$$A_0 = \frac{1}{4} \quad \text{and} \quad A_m = \begin{cases} \frac{6}{(\pi m)^2} & \text{if } m \text{ is even} \\ -\frac{6}{(\pi m)^2} + \frac{24}{(\pi m)^4} & \text{if } m \text{ is odd} \end{cases}$$

Application: You can now go back and solve the wave equation problem that we discussed in the separation of variables lecture:

Example 2: (more practice)

$$\begin{cases} u_{tt} = c^2 u_{xx} \\ u_x(0, t) = 0 \\ u_x(1, t) = 0 \\ u(x, 0) = x^3 \\ u_t(x, 0) = \cos(3\pi x) \end{cases}$$

PARTS I–V: From the Separation of Variables lecture we got

$$u(x, t) = (A_0 + B_0 t) + \sum_{m=1}^{\infty} (A_m \cos(\pi m c t) + B_m \sin(\pi m c t)) \cos(\pi m x)$$

PART VI: INITIAL CONDITION

$$u(x, 0) = A_0 + B_0(0) + \sum_{m=1}^{\infty} (A_m \cos(\pi m c 0) + B_m \sin(\pi m c 0)) \cos(\pi m x)$$

$$x^3 = A_0 + \sum_{m=1}^{\infty} A_m \cos(\pi m x) = A_0 \cos(\pi 0 x) + \sum_{m=1}^{\infty} A_m \cos(\pi m x)$$

$$x^3 = \sum_{m=0}^{\infty} A_m \cos(\pi m x)$$

Insert Previous Example

$$A_0 = \frac{1}{4} \quad \text{and} \quad A_m = \begin{cases} \frac{6}{(\pi m)^2} & \text{if } m \text{ is even} \\ -\frac{6}{(\pi m)^2} + \frac{24}{(\pi m)^4} & \text{if } m \text{ is odd} \end{cases}$$

PART VII: INITIAL VELOCITY: We previously obtained

$$B_0 = 0 \text{ and } B_1 = 0 \text{ and } B_2 = 0 \text{ and } B_3 = \frac{1}{3\pi c} \text{ and } B_4 = 0 \dots$$

PART VIII: Solution

$$u(x, t) = (A_0 + B_0 t) + \sum_{m=1}^{\infty} (A_m \cos(\pi m c t) + B_m \sin(\pi m c t)) \cos(\pi m x)$$

$$\text{Where } A_0 = \frac{1}{4} \text{ and } B_0 = 0$$

$$\text{And } A_m = \begin{cases} \frac{6}{(\pi m)^2} & \text{if } m \text{ is even} \\ -\frac{6}{(\pi m)^2} + \frac{24}{(\pi m)^4} & \text{if } m \text{ is odd} \end{cases} \quad \text{and } B_m = \begin{cases} 0 & \text{if } m \neq 3 \\ \frac{1}{3\pi c} & \text{if } m = 3 \end{cases}$$

3. FULL FOURIER SERIES

Video: Full Fourier Series

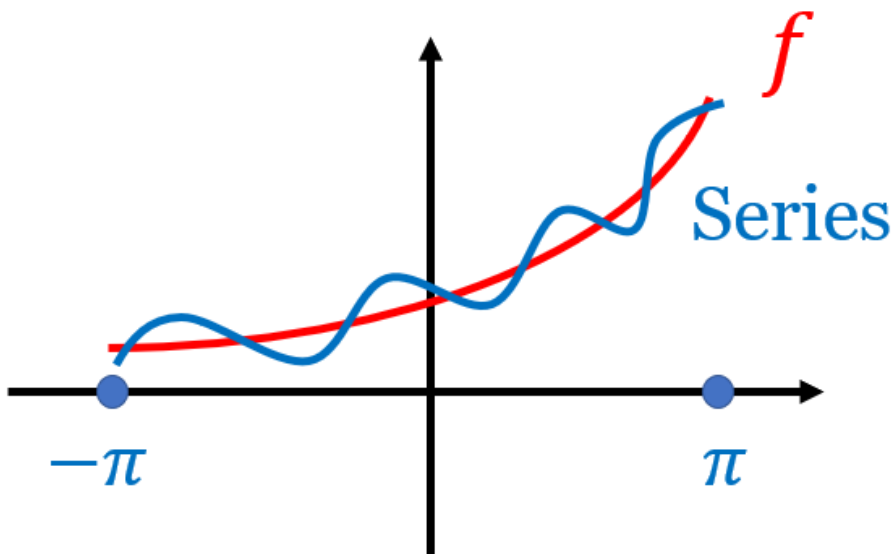
Question: What if you want to write f in terms of sin **and** cos?

Goal:

Find A_m and B_m such that

$$f(x) = \sum_{m=0}^{\infty} A_m \cos(mx) + B_m \sin(mx) \text{ on } (-\pi, \pi)$$

Notice the bigger interval here!



Everything we said so far is still true, except our dot product is

$$f \cdot g = \int_{-\pi}^{\pi} f(x)g(x)dx$$

And in particular we get

$$A_m = \frac{f \cdot \cos(mx)}{\cos(mx) \cdot \cos(mx)} = \frac{\int_{-\pi}^{\pi} f(x) \cos(mx) dx}{\int_{-\pi}^{\pi} \cos^2(mx) dx} = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(mx) dx$$

$$B_m = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(mx) dx$$

Note: This time the denominator is π and not $\frac{\pi}{2}$ because we're working with the **full** interval $(-\pi, \pi)$

Special Case: $m = 0$

$$A_0 = \frac{f \cdot \cos(0x)}{\cos(0x) \cdot \cos(0x)} = \frac{\int_{-\pi}^{\pi} f(x) dx}{\int_{-\pi}^{\pi} 1 dx} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$B_0 = 0 \quad (\text{By convention, since it corresponds to } \sin(0x) = 0)$$

Full Fourier Series:

If $f(x) = \sum_{m=0}^{\infty} A_m \cos\left(\frac{\pi m x}{L}\right) + B_m \sin\left(\frac{\pi m x}{L}\right)$ on $(-L, L)$ then

$$A_0 = \frac{1}{2L} \int_{-L}^L f(x) dx \text{ and } B_0 = 0 \text{ and}$$

$$A_m = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{\pi m x}{L}\right) dx \text{ and}$$

$$B_m = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{\pi m x}{L}\right) dx$$

Example 3:

Find A_m and B_m such that we have

$$x = \sum_{m=0}^{\infty} A_m \cos(mx) + B_m \sin(mx) \text{ on } (-\pi, \pi)$$

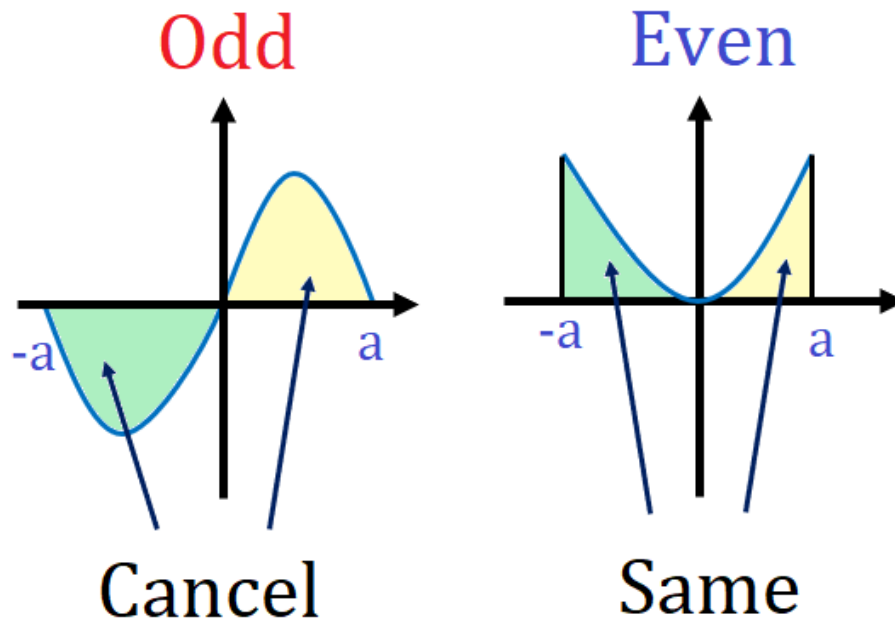
Special Case: $m = 0$

$$A_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x dx = \frac{0}{2\pi} = 0 \text{ and } B_0 = 0$$

General Case: $m = 1, 2, \dots$

$$A_m = \frac{1}{\pi} \int_{-\pi}^{\pi} \underbrace{x \cos(mx)}_{\text{Odd}} dx = 0$$

$$B_m = \frac{1}{\pi} \int_{-\pi}^{\pi} \underbrace{x \sin(mx)}_{\text{Even}} dx = \frac{\textcolor{red}{2}}{\pi} \int_{\textcolor{red}{0}}^{\pi} x \sin(mx) dx = \left(\frac{2}{m} \right) (-1)^{m+1} \text{ (From last time)}$$



Therefore we get

$$x = 0 \cos(0x) + 0 \sin(0x) + \sum_{m=1}^{\infty} 0 \cos(mx) + \left(\frac{2}{m}\right) (-1)^{m+1} \sin(mx)$$

Which simplifies to

$$x = \sum_{m=1}^{\infty} \left(\frac{2}{m}\right) (-1)^{m+1} \sin(mx) \quad (\text{Sine Series!})$$

Note: In fact, this generalizes: If f is odd on **the full interval** $(-\pi, \pi)$ then the Full Fourier series is a sine series, and similar if f is even

Note: But in general, you really need to calculate **both** A_m and B_m